

ORANGE COUNTY AP, NY, USA

WMO: 725015

Lat: 41.509N

Lon: 74.265W

Elev: 111

StdP: 100.00

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04789

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.1	-22.6	0.5	-14.5	-20.2	0.6	-11.3	10.8	-4.8	9.3	-3.4	2.1	300	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.6	23.2	31.1	22.6	29.4	21.6	24.7	30.0	23.9	28.8	23.0	27.6	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.0	27.1	22.4	17.3	26.4	21.7	16.6	25.6	75.2	30.0	72.0	28.7	68.9	27.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.9	7.9	7.1	DB	-20.2	35.4	3.7	2.5	-22.9	37.2	-25.0	38.7	-27.1	40.1	-29.8	41.9	(4)
(5)			WB	-20.4	26.2	3.6	1.0	-23.0	27.0	-25.1	27.5	-27.1	28.1	-29.7	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.2	-1.9	3.0	9.5	15.2	20.0	22.7	21.7	17.7	11.2	5.3	-0.1	(6)
(7)		DBStd	10.01	5.66	5.16	5.13	4.72	4.19	3.40	2.88	2.94	3.82	4.40	4.51	4.80	(7)
(8)		HDD10.0	1541	409	333	226	64	4	0	0	0	1	38	152	313	(8)
(9)		HDD18.3	3365	667	566	477	268	117	22	2	6	57	224	391	570	(9)
(10)		CDD10.0	1599	1	1	7	49	166	299	392	363	232	77	11	1	(10)
(11)		CDD18.3	383	0	0	0	3	20	71	136	111	39	4	0	0	(11)
(12)		CDH23.3	3628	0	0	2	69	258	680	1292	947	344	36	0	0	(12)
(13)		CDH26.7	1193	0	0	0	20	77	208	482	306	94	5	0	0	(13)
(14)	Wind	WSAvg	2.7	3.1	3.1	3.3	3.2	2.6	2.4	2.1	1.9	2.0	2.5	2.7	2.9	(14)
(15)	Precipitation	PrecAvg	1111	77	62	86	93	107	109	100	96	102	90	95	81	(15)
(16)		PrecMax	1496	250	116	192	224	258	291	346	243	251	455	194	190	(16)
(17)		PrecMin	719	4	6	10	30	30	10	21	16	19	1	21	21	(17)
(18)		PrecStd	194	45	26	38	38	54	60	59	50	60	70	43	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	15.9	21.6	29.5	31.6	32.9	35.0	34.8	32.3	27.0	21.0	16.6	(19)
(20)			MCWB	11.8	11.9	13.6	17.2	21.2	23.5	24.2	24.0	23.1	19.5	15.9	13.3	(20)
(21)		2%	DB	10.2	11.5	17.7	24.9	28.9	31.1	32.9	32.0	29.3	23.8	17.9	12.5	(21)
(22)			MCWB	7.5	7.4	11.6	15.2	20.0	22.9	23.6	23.3	21.5	17.8	14.0	10.1	(22)
(23)		5%	DB	6.9	7.8	14.1	21.7	26.4	29.0	31.2	29.8	27.3	21.4	15.8	9.6	(23)
(24)			MCWB	4.3	4.3	8.5	13.7	18.7	21.4	22.9	22.1	20.4	17.1	11.8	7.0	(24)
(25)		10%	DB	4.4	5.4	11.3	18.5	23.6	27.4	29.3	28.1	25.1	19.1	13.0	7.2	(25)
(26)			MCWB	2.0	2.4	7.2	11.7	17.0	20.6	21.9	21.2	19.5	15.1	9.6	4.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	13.3	15.2	19.1	23.3	24.9	26.0	25.8	24.5	21.6	17.3	14.4	(27)
(28)			MCDWB	14.0	15.1	19.4	26.2	29.0	30.3	32.6	31.7	29.2	24.4	19.4	15.7	(28)
(29)		2%	WB	7.8	8.0	12.3	16.9	21.4	23.8	24.8	24.4	23.1	19.6	14.9	10.6	(29)
(30)			MCDWB	9.6	11.2	16.5	22.8	26.8	29.0	30.4	29.3	27.3	22.7	17.4	12.3	(30)
(31)		5%	WB	4.6	4.8	9.6	14.9	19.9	22.8	24.0	23.5	21.8	17.7	12.5	7.7	(31)
(32)			MCDWB	6.5	7.6	13.3	20.3	24.6	27.4	29.0	28.1	25.5	20.7	14.6	9.2	(32)
(33)		10%	WB	2.5	2.5	7.4	12.6	18.4	21.7	23.0	22.5	20.7	15.6	10.4	5.0	(33)
(34)			MCDWB	4.3	4.9	10.8	17.1	22.3	25.8	27.5	26.6	23.9	18.5	12.6	6.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	10.2	10.6	12.5	12.4	11.9	12.1	11.8	12.3	11.6	10.7	8.6	(35)
(36)			MCDBR	11.5	13.6	15.7	18.1	16.1	14.3	13.9	13.6	14.4	14.7	14.1	11.4	(36)
(37)			MCWBR	9.4	10.2	9.9	9.6	8.5	6.9	5.9	6.0	7.4	9.1	10.1	9.5	(37)
(38)		5% WB	MCDWR	11.2	12.8	14.4	15.9	13.7	12.5	11.9	11.5	11.6	12.4	12.0	10.4	(38)
(39)			MCWBR	9.7	10.2	10.0	9.5	7.9	6.6	5.6	5.7	6.6	8.8	9.9	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.321	0.347	0.363	0.406	0.467	0.474	0.455	0.383	0.355	0.320	0.302	(40)	
(41)		taud	2.414	2.369	2.384	2.391	2.296	2.160	2.152	2.197	2.374	2.461	2.514	2.495	(41)	
(42)		Ebn at Noon	851	885	901	907	871	816	804	809	851	837	826	823	(42)	
(43)		Edh at Noon	79	97	107	115	129	149	149	138	108	88	72	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.76	2.64	3.62	4.60	5.14	5.57	5.80	5.07	4.21	2.82	1.95	1.44	(44)	
(45)		RadStd	0.13	0.26	0.28	0.49	0.52	0.53	0.33	0.33	0.46	0.31	0.19	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+61

Nomenclature: See separate page